```
# Mount Google Drive
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
Wcxn_input = "/content/drive/MyDrive/RevEngg_Qwen_Gemini/Augmented_code_Qwen_Chatgpt_Gemini_summary_similarity_numeric.csv"
```

```
import pandas as pd
wcxn = pd.read_csv(Wcxn_input)
```

```
wcxn.columns
```

```
Index(['Name_of_file', 'Qwen_Similarity_Score', 'Chatgpt_Similarity_Score',
       'Gemini_Similarity_Score', 'Score_verilog', 'SimilarityChecker_Qwen',
       'SimilarityChecker_Gemini', 'SimilarityChecker_Chatgpt'],
      dtype='object')
```

Step 1: Understanding Wilcoxon Signed-Rank Test

Null hypothesis (H_0): The two paired samples come from the same distribution (no significant difference).

Alternative hypothesis (H_1): The two paired samples differ significantly.

p-value < 0.05 ⟶ Reject H_0 ⟶ Significant difference

p-value ≥ 0.05 ⟶ Fail to reject H_0 ⟶ No significant difference (they are similar)

Statistic:

A small W (statistic) ⟶ large difference between paired samples ⟶ groups differ.

A large W (close to total ranks) ⟶ small difference ⟶ groups are similar.

The p-value tells you if that difference is statistically significant.*italicised text*

```
import itertools
from scipy.stats import wilcoxon

# Select only numeric columns (Wilcoxon requires numeric data)
numeric_cols = wcxn.select_dtypes(include=['number']).columns

results = []

# Iterate over all pairs of numeric columns
for col1, col2 in itertools.combinations(numeric_cols, 2):
    stat, p = wilcoxon(wcxn[col1], wcxn[col2])
    results.append({
        'Column 1': col1,
        'Column 2': col2,
        'Statistic': stat,
        'p-value': p
    })

wilcoxon_results = pd.DataFrame(results)

# print(wilcoxon_results)

sorted_results = wilcoxon_results.sort_values(
    by=["p-value", "Statistic"],
    ascending=[True, True]   # first by p-value (small → large), then by Statistic (small → large)
)
print(sorted_results)
```

	Column 1	Column 2	Statistic	p-value
15	Score_verilog	SimilarityChecker_Qwen	0.0	0.000002
3	Qwen_Similarity_Score	SimilarityChecker_Qwen	0.0	0.000002
12	Gemini_Similarity_Score	SimilarityChecker_Qwen	0.0	0.000002
8	Chatgpt_Similarity_Score	SimilarityChecker_Qwen	0.0	0.000002
17	Score_verilog	SimilarityChecker_Chatgpt	1.0	0.000002
5	Qwen_Similarity_Score	SimilarityChecker_Chatgpt	1.0	0.000002
10	Chatgpt_Similarity_Score	SimilarityChecker_Chatgpt	1.0	0.000002
14	Gemini_Similarity_Score	SimilarityChecker_Chatgpt	1.0	0.000002
13	Gemini_Similarity_Score	SimilarityChecker_Gemini	3.0	0.000002
4	Qwen_Similarity_Score	SimilarityChecker_Gemini	3.0	0.000002
9	Chatgpt_Similarity_Score	SimilarityChecker_Gemini	3.0	0.000002
16	Score_verilog	SimilarityChecker_Gemini	3.0	0.000002
2	Qwen_Similarity_Score	Score_verilog	27.0	0.002741
7	Chatgpt_Similarity_Score	Score_verilog	55.0	0.009647
19	SimilarityChecker_Qwen	SimilarityChecker_Chatgpt	24.5	0.013647
20	SimilarityChecker_Gemini	SimilarityChecker_Chatgpt	115.5	0.046118
11	Gemini_Similarity_Score	Score_verilog	32.0	0.052260
6	Chatgpt_Similarity_Score	Gemini_Similarity_Score	60.0	0.079696
0	Qwen_Similarity_Score	Chatgpt_Similarity_Score	41.5	0.153812
1	Qwen_Similarity_Score	Gemini_Similarity_Score	20.0	0.405381
18	SimilarityChecker_Qwen	SimilarityChecker_Gemini	166.5	0.588419

Start coding or generate with AI.

Explanation of Each Column

Column Name	Description
Name_of_file	The name of the file under evaluation.
Qwen_Similarity_Score	The similarity score (1–5) given **by Gemini** for how well the **Qwen-predicted code** matches the **original C code summary**.
Chatgpt_Similarity_Score	The similarity score (1–5) given **by Gemini** for how well the **ChatGPT-predicted code** matches the **original C code summary**.
Gemini_Similarity_Score	The similarity score (1–5) given **by Gemini** for how well **Gemini's own predicted code** matches the **original C code summary**.
Score_verilog	A ranking (1–5) given **by Gemini** evaluating how closely the **Verilog code** corresponds to the **original C code**, in terms of logic and functionality.
SimilarityChecker_Qwen	The similarity score computed by the **Sim_C model (Similarity Checker)** for the **Qwen-predicted code** compared with the **original C code**.
SimilarityChecker_Gemini	The similarity score computed by the **Sim_C model (Similarity Checker)** for the **Gemini-predicted code** compared with the **original C code**.
SimilarityChecker_Chatgpt	The similarity score computed by the **Sim_C model (Similarity Checker)** for the **ChatGPT-predicted code** compared with the **original C code**.

Start coding or generate with AI.